Nirav Sai Shah
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Senior Data Analyst

Summary

- Experienced data analyst with 8 years of expertise in financial data analysis, risk management, and regulatory compliance within the banking sector with expertise in tools like SQL, Tableau, Power BI, and SAS.
- Proficient in utilizing SQL to query, manipulate, and analyze large datasets across various relational databases like Oracle and SQL Server for efficient data extraction and reporting.
- Proven experience working in Agile environments, collaborating with cross-functional teams to deliver data-driven solutions efficiently and adapt to evolving business requirements, ensuring timely project completion and high-quality results.
- Expert in defining, tracking, and analyzing key performance indicators (KPIs) to measure business performance, drive data-informed decisions, and deliver actionable insights that support strategic goals and operational efficiency.
- Skilled in analyzing large datasets within core banking systems like Temenos and Finacle, using SQL, Python, and Excel to identify risks, optimize operations, and deliver actionable business insights.
- Skilled in analyzing transaction data and identifying suspicious activities using AML systems like Actimize and SAS AML, ensuring compliance with Anti-Money Laundering regulations and Know Your Customer (KYC) requirements.
- Expertise in financial risk assessment, working with tools like SAS Risk Management and Moody's Analytics to mitigate market, credit, and operational risks.
- Skilled in using Tableau and Power BI to create dashboards, ensuring data security and compliance with GDPR, while using Python, R, and Excel to analyze financial trends and provide insights that drive business decisions in the banking sector.
- Adept at performing predictive analysis using Python and R, leveraging machine learning libraries like Scikit-learn and Pandas to build models for customer behavior prediction and loan default forecasting.
- Proficient in ETL processes using tools such as Informatica and Talend, streamlining the extraction, transformation, and loading of financial data into centralized data warehouses.
- Skilled in analyzing data from banking systems like Flex cube and creating reports using SAP and OFSAA to support business strategy and decision-making.
- Extensive experience working with payment systems like SWIFT and RTGS, ensuring accurate transaction processing and identifying anomalies for fraud prevention.
- Expertise in leveraging CRM tools like Salesforce to analyze customer data, improve customer segmentation, and drive targeted marketing strategies in the banking sector.
- Adept at applying statistical analysis methods using Excel, SAS, and R to assess financial data trends, test hypotheses, and derive insights for informed decision-making.
- Experienced data analyst focused on regulatory compliance and reporting in banking.
- Skilled in creating accurate reports using tools like Tableau BI, and ensuring compliance with regulations such as Basel III and GDPR using data from core banking systems like Temenos and Finacle.
- Skilled in using data visualization tools like QlikView and Power BI to create real-time reporting solutions that enhance business intelligence and operational efficiencies.
- Proficient in managing financial data across multiple platforms, utilizing tools such as Google Big Query and Amazon Redshift to analyze large-scale transactional datasets.
- Expertise in conducting credit risk assessments using tools like FICO and Experian, analyzing credit scores, and building predictive models for loan default risk.
- Experience in preparing regulatory reports and compliance documentation using AxiomSL and Wolters Kluwer, ensuring all banking activities meet statutory requirements.
- Strong analytical abilities in forecasting financial trends, creating performance metrics, and providing actionable insights using Python, R, and Excel to drive business outcomes in the banking industry.
- Proficient in Master Data Management (MDM), maintaining data consistency across key banking systems like Core Banking Platforms, ensuring the integrity of customer and transaction data.

- Proficient in data governance and validation within the banking industry, ensuring compliance with regulations like GDPR, Basel III, and Dodd-Frank while maintaining data accuracy for regulatory reporting.
- Experienced in working with Databricks to process and analyze large-scale banking data, enabling efficient automation of data workflows and enhanced data processing performance.
- Skilled in building data models to analyze banking data trends, providing deeper insights into customer behavior, loan defaults, and risk management.
- Expertise in data mapping to ensure data consistency and integration across different banking systems, ensuring alignment of financial data for regulatory and reporting purposes.
- Strong knowledge of data lineage to track the flow of banking data across systems, ensuring full traceability and transparency in financial transactions and regulatory compliance.
- Proficient in data integration, consolidating financial data from core banking systems and payment platforms like SWIFT to create comprehensive datasets for reporting and analysis.
- Skilled in using big data tools like Hadoop and Spark for processing large volumes of banking data, identifying patterns for market analysis, customer segmentation, and fraud detection.

Professional Experience

Sr Data Analyst
JPMorgan Chase & Co –(New York) **Oct 2022- Present**

- Collected and integrated diverse data sources, including customer onboarding (KYC), transaction logs, and historical data from structured (SQL databases) and unstructured (logs, emails) sources, ensuring seamless data flow for the transaction monitoring system.
- Ensured that all data collection and integration processes adhered to the bank's data governance framework, including compliance with regulations such as GDPR, Data Protection Act, and Basel III for financial data privacy and security,.
- Developed credit risk prediction models using Python (scikit-learn, pandas) ,Excel and SQL, leveraging Logistic Regression and Random Forest to assess default probabilities, improving risk classification accuracy by 20%.
- Collaborated with data governance teams to implement security measures, access controls, and data encryption protocols on sensitive customer and transaction data, ensuring adherence to data privacy and compliance regulations.
- Implemented a scalable monitoring system using Apache Hadoop and Apache Spark to process large volumes of customer transaction data in real-time.
- Led data quality and cleansing efforts, implementing automated checks for data consistency, completeness, and accuracy using Python and R ensuring high-quality data for AML/KYC monitoring systems.
- Developed and maintained automated workflows for detecting and correcting missing values, outliers, and duplicate data, improving data integrity for real-time fraud detection.
- Established and maintained comprehensive data lineage processes, ensuring all transaction data was traceable with audit trails for regulatory reporting and compliance with standards like FATF and SOX.
- Enforced data retention policies for archiving and disposing of sensitive data, complying with financial regulations such as MiFID II and GDPR, ensuring the safe and compliant handling of customer information.
- Collaborated with data science teams to develop and refine risk scoring models, scoring transactions based on customer behavior, geographical location, transaction amount, and history, for identifying potential money laundering activities.
- Built fraud prediction models using Random Forest and Logistic Regression in Python, improving AML accuracy and aligning with FATF monitoring guidelines.
- Designed and deployed interactive dashboards using Power BI to visualize real-time transaction data and provide actionable insights for compliance officers and management.
- Developed automated KYC compliance reports based on customer onboarding data, ensuring the system met legal requirements for identity verification and reducing the risk of fraud.
- Collaborated with finance and accounting teams to integrate AML and KYC data with existing financial reporting systems (e.g., SAP, Oracle) for a comprehensive view of risk-related financial metrics.

- Implemented automated alerting systems for flagging high-risk transactions based on pre-set thresholds (e.g., large transactions, rapid movement of funds), improving the detection of suspicious behavior.
- Automated the generation of detailed compliance reports, using Python scripts, ensuring timely submission to regulatory authorities like the FCA and FINCEN.
- Developed real-time transaction monitoring workflows using Apache Kafka and AWS Lambda, processing high-frequency transaction streams to identify suspicious patterns and ensure AML compliance.
- Ensured that all financial reports, including those related to AML and KYC, complied with CCPA data privacy standards, preventing unauthorized access and protecting customer confidentiality.
- Worked with the risk management and compliance teams to ensure all monitoring activities aligned with the bank's internal risk management framework and global regulatory standards.
- Created models to organize customer and transaction data, focusing on details like transaction size, frequency, and location. This helped in building accurate risk scores to quickly detect suspicious activities using Python
- Mapped data from different sources (like customer info and transaction logs) to make sure it worked smoothly with the AML/KYC system. This helped ensure the data was properly organized for accurate reporting and risk checks using Python.
- Played a key role in preparing for and supporting internal and external compliance audits by ensuring that all transactional data, risk models, and reports met regulatory standards.
- Generated comprehensive reports on financial risks tied to AML and KYC violations, helping the bank to track non-compliant transactions and identify high-risk customer profiles.
- Coordinated with senior management and compliance officers to present detailed insights on suspicious activities, financial risks, and regulatory compliance, aiding in timely decision-making.

Data Analyst
Goldman Sachs(Houston- Texas) **August-2020- Sept 2022**

- Cleaned and validated CRM (e.g., Salesforce) and Risk Management (e.g., SAS Risk Management) data for AML, KYC, and GDPR compliance. Used SQL for real-time transaction and account monitoring.
- Implemented data lineage solutions with Apache Atlas and Informatica to track CRM and Risk Management data. Ensured PCI DSS and GDPR compliance by documenting transformations and maintaining traceability.
- Developed automated compliance reports with Power BI to monitor suspicious activities. Automated KYC and fraud data extraction for AML/KYC regulation compliance.
- Integrated CRM (e.g., Salesforce) and Risk Management (e.g., SAS Risk Management) data into financial reports using SAP . Automated reports aligned with IFRS and GAAP for fraud risk transparency.
- Created real-time compliance dashboards using Power BI to visualize suspicious activities. Developed interactive dashboards to track AML, KYC, and regulatory risks.
- Integrated CRM (e.g., Salesforce), Risk Management, and Fraud Detection data with Apache Kafka and AWS Lambda. Built a cross-system data pipeline for early fraud and AML violation detection.
- Monitored compliance risks across CRM (e.g., Salesforce) and Risk Management systems using Python and SQL. Automated risk assessment to flag non-compliant activities in real-time.
- Developed data governance protocols with Informatica to meet GDPR and PCI DSS standards. Implemented encryption and anonymization using Azure Key Vault for secure data integration.
- Created an audit trail system with Apache Atlas to track data transformations. Ensured transparency for audits and compliance with KYC and AML regulations.
- Supported internal audits by documenting data processes in CRM (e.g., Salesforce) and Risk Management (e.g., SAS Risk Management). Assisted auditors in compliance with IFRS, GAAP, AML, and KYC.
- Designed a data governance framework using Informatica to track data changes for compliance. Created documentation for every data change, ensuring GDPR compliance.
- Implemented real-time monitoring with Apache Kafka and AWS Lambda to ensure continuous AML compliance. Used Tableau and Power BI for automated alerts and dashboards.
- Created dynamic compliance reports in SQL, based on data from CRM and Risk Management systems. Integrated Pandas and NumPy for real-time KYC and AML monitoring.

- Leveraged Azure ML and Python to implement proactive monitoring of KYC breaches and generate early alerts for potential AML violations.
- Automated transaction legitimacy monitoring across CRM (e.g., Salesforce) and Risk Management systems using SQL. Developed fraud detection models with Scikit-learn for real-time alerts.
- Developed predictive fraud detection models using Python and Scikit-learn TensorFlow/Keras use in AML was exploratory — consider generalizing to ensure realism.
- Tracked customer transactions using Power BI for AML compliance. Maintained data documentation with Informatica for regulatory reporting transparency.
- Integrated fraud risk indicators from SAS Fraud Management into compliance reports using Power BI. Monitored fraud risks across CRM and Risk Management systems for AML compliance.
- Cleaned and mapped data from Salesforce and SAS Risk Management for AML and KYC compliance using Python for real-time monitoring.
- Integrated data from Salesforce, SAS Risk Management, and fraud detection systems into reports with SAP and Oracle, ensuring compliance with IFRS and GAAP.
- Collaborated with stakeholders to present compliance insights using Power BI and Tableau. Delivered regular updates on AML, KYC, and Risk Management compliance.
- Ensured customer data protection in line with GDPR and PCI DSS regulations. Managed sensitive data during reporting using Azure Key Vault and AWS KMS for secure data handling.

Data Analyst
IBM(San Francisco, CA) **Jun 2018- July 2020**

- Integrated customer data from core banking and trading platforms using Python for enhanced behavior analysis. Used ETL tools like Apache NiFi to integrate data across systems.
- Analyzed transaction history and demographics from core banking to segment customers using Python and clustering algorithms. Applied Scikit-learn to identify high-value segments.
- Applied **decision trees and isolation forests** to detect anomalies in transaction behavior, identifying potential fraud based on customer transaction history.
- Integrated financial crime detection systems with core banking data using Apache Kafka for real-time monitoring. Developed automated alerts in SQL for high-risk transactions.
- Merged data from core banking and trading systems using Python to generate insights into customer behavior. Analyzed data to uncover cross-selling opportunities using Pandas and NumPy.
- Built customer recommendation models using Python (Scikit-learn)**,** analyzing behavior data from core banking systems to suggest suitable financial products.
- Ensured IFRS compliance in reports on customer activities and fraud detection. Collaborated with finance teams to maintain transparency using SQL
- Ensured GDPR, PCI DSS, and AML compliance by tracking data lineage with Informatica. Implemented data anonymization using AWS KMS and Azure Key Vault.
- Built machine learning models using Python to assess fraud risk based on customer transaction history. Integrated fraud patterns from SAS Fraud Management.
- Ensured data accuracy and consistency using Pandas and NumPy for cleaning and validation. Followed data governance standards to ensure high-quality data for decision-making.
- Created interactive dashboards using Power BI to visualize customer metrics and fraud risks. Enabled relationship managers to proactively address fraud risks.
- Collaborated with risk management and fraud teams using SQL to integrate insights. Used Power BI to drive customer engagement strategies.
- Ensured adherence to data security and regulatory standards like KYC, AML in handling sensitive customer data. Implemented access control protocols using Azure Active Directory and AWS IAM.
- Tailored financial products based on customer behavior using core banking and trading data. Applied machine learning models in Python to recommend personalized products.
- Maintained audit trails for all data analysis to ensure compliance with AML and KYC regulations. Conducted audits using Splunk and Tableau to document fraud detection changes.

- Used insights from fraud detection and core banking data to enhance financial crime detection. Deployed real-time fraud models in Python to detect anomalies.
- Tracked KPIs for customer behavior and fraud detection using Python and Power BI to drive data-driven decisions and improve fraud risk management.
- Worked with finance teams to assess the financial impact of fraud and generate IFRS-compliant reports. Documented financial consequences using SQL for regulatory reporting.
- Analyzed customer activities across core banking and trading systems to build risk profiles. Applied SQL and machine learning models like decision trees and SVM for fraud detection.
- Ensured ethical data use in fraud detection and product recommendations. Documented model assumptions and validation techniques for GDPR compliance using Python and Jupiter Notebooks.
- Integrated customer transaction data from core banking and trading platforms using Apache NiFi, enabling seamless analysis for behavior and fraud detection.
- Applied Scikit-learn for customer segmentation and behavior modeling, using transaction and demographic data to identify high-value segments and potential fraud risks.
- Developed a real-time fraud monitoring system integrating core banking and financial crime detection systems. Enabled proactive fraud detection with automated alerts using Apache Kafka.

Junior Data Analyst
Bank of America(Tampa- Florida) **April-2016- May 2018**

- Assisted in gathering payment and customer data from systems like Salesforce and SWIFT to support AML and KYC compliance reporting.
- Helped automate compliance reporting processes using SQL and Python, ensuring timely updates based on SWIFT payment data.
- Supported the development of Power BI dashboards to display AML compliance metrics; assisted in updating data using Python and SQL.
- Cleaned and validated sensitive customer data using SQL and Python, following internal data governance guidelines.
- Helped combine data from payment systems, CRM, and trading platforms using SQL and Python for regulatory reporting.
- Performed data checks using Python (Pandas) to ensure accurate AML and KYC compliance data from various banking systems.
- Supported data integration efforts for payment transactions and customer data to help meet AML/KYC requirements.
- Used SQL and Python to prepare compliance data for reporting, helping maintain accuracy and completeness.
- Assisted in preparing Power BI reports aligned with financial standards, using SQL to extract data and support compliance needs.
- Helped automate regular KYC/AML reports using SQL queries and Python scripts, improving report turnaround times.
- Maintained basic audit logs using SQL to support transparency and traceability of compliance data.
- Assisted in reviewing transaction data to help identify potentially suspicious activities using SQL and Python.
- Ensured that all changes to data were recorded for audit purposes using SQL and documentation tools.
- Supported compliance teams by providing transaction data and summaries for identifying high-risk behavior patterns.
- Reconciled data from different banking systems using SQL to ensure accurate and complete compliance reports.
- Helped automate steps in the compliance reporting process using Python and created basic visual reports in Tableau
- Followed data privacy best practices while working with customer data to support GDPR and PCI DSS compliance.
- Helped set up alerts in SQL and Power BI to assist compliance teams in monitoring unusual transactions.
- Used SQL and Tableau to assist in streamlining compliance reports and improving report accuracy.